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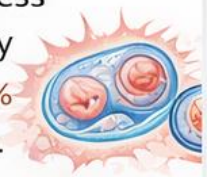
Project 1: Pan-Stress Forces Loss of Stemness and Drives Developmental Imbalance

Background

Environmental stress contributes to early miscarriages in 70% of human embryos.



Objective

Identify Common "Pan-Stress" Response Genes in ESCs

Method

Cell Culture
↓
Bulk RNAseq
↓
scRNAseq

KEY FINDINGS

1. **All stressors override stemness, forcing ESCs out of their naïve (0th) lineage state**
2. **Stress responses segregate into two distinct lineage-imbalance trajectories**
3. **Identified two pan-stress marker genes that robustly predict lineage imbalance** independent of stressor identity

Publications & IP:

- Scientific Reports (2025)
- Patent filed April 2025
- Birth Defects Research (2024)

Project 2: scRNA-seq Identification of tissue-resident DCs in Endometrium

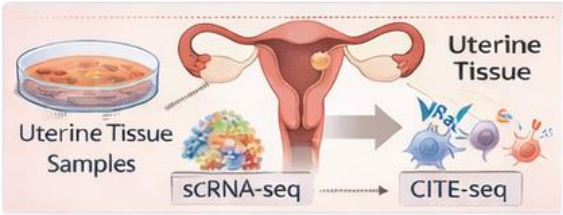
Problem

DC depletion causes **Implantation failure**

Objective

Map the Developmental Trajectory of uDCs

Approach



KEY FINDINGS

1. Identified 7 subsets of **uDCs including progenitor cells** thus challenging the dogma that DCs are recruited from peripheral blood.
2. **Mapped the developmental trajectory** of uDCs through the various stages of endometrial cycle.
3. Assigned **functional roles** to each subset based on their gene expression profiles.

Functional Roles

Publication:

- iScience (2025)

Project 3: Sex specific Immune Regulation at the Maternal-Fetal Interface

Background

Fetal sex strongly influences placental and maternal immune responses

Objective

Understand how fetal sex programs immune architecture during pregnancy

Approach

Bulk RNAseq

↓

Single-cell RNAseq

↓

CellChat

KEY FINDINGS

1. **Sex-specific gene expression** differences emerge early in normal placental development
2. **Male placentas show innate inflammatory skewing**; females favored adaptive immune regulation
3. Distinct immune failure pathways identified in miscarriages

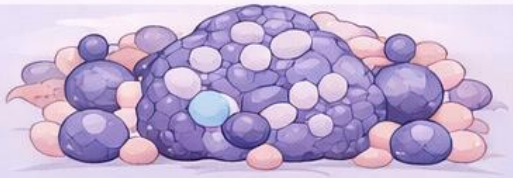
Publication:

- Sex-specific Immune *Regulation at the Maternal-Fetal Interface*, [manuscript under review] 2026

Project 4: Cancer Cell Sialylation Fuels Immune Evasion and Stromal Remodeling in Ovarian Cancer

Problem

Tumor cell hypersialylation reprograms the ovarian tumor microenvironment to suppress anti-tumor **immunity**



Immune Suppressive Tumor Microenvironment

Objective

Investigate How **Hypersialylation** Drives Immune Suppression in Ovarian Cancer

Immune Suppressive Tumor Microenvironment

Approach

CRISPR KO

↓

Flow Cytometry

↓

Bulk RNAseq

↓

scRNAseq

KEY FINDINGS

1. **Tumor cell hypersialylation** drives immune-dependent ovarian cancer progression.
2. **ST6GAL1-mediated sialylation** promotes expansion of immunosuppressive myeloid cells and suppresses cytotoxic immunity.
3. **Hypersialylation** impairs innate danger sensing and reprograms the tumor microenvironment.
4. **Targeting sialylation** reverses immune suppression and prevents tumor progression.